

1. $u, v \in V, \langle \cdot, \cdot \rangle, \tilde{u}, \tilde{v} \in V$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\begin{aligned} \text{Dk: } \|u+v\|^2 &\stackrel{\text{SLZ}}{=} \langle u+v, u+v \rangle = \langle u+v, u \rangle + \langle u+v, v \rangle \\ &= \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle \\ &\stackrel{\text{SLZ}}{=} \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} \end{aligned}$$

$$\begin{aligned} \|u-v\|^2 &\stackrel{\text{SLZ}}{=} \langle u-v, u-v \rangle = \langle u-v, u \rangle - \langle u-v, v \rangle \\ &= \langle u, u \rangle - \langle v, u \rangle - (\langle u, v \rangle - \langle v, v \rangle) \\ &\stackrel{\text{SLZ}}{=} \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \overline{\langle u, v \rangle} \end{aligned}$$

Při úpravách jsme použili vlastnosti skalárního součinu a poznámku 7.15

Potom

$$\begin{aligned} \|u+v\|^2 - \|u-v\|^2 &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} \\ &\quad - \|u\|^2 - \|v\|^2 + \langle u, v \rangle - \overline{\langle u, v \rangle} \\ &= 2\langle u, v \rangle + 2\overline{\langle u, v \rangle} = 4\operatorname{Re}(\langle u, v \rangle) \end{aligned}$$

$$\text{Pomůžeme } z + \bar{z} = 2\operatorname{Re}(z)$$

Uvedeme-li a, b komplexní

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) \quad \bullet$$

Ověřme analogický případ

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\begin{aligned} \|u-iv\|^2 &\stackrel{\text{SLZ}}{=} \langle u-iv, u-iv \rangle = \langle u-iv, u \rangle - i\langle u-iv, v \rangle \\ &= \langle u, u \rangle + \langle -iv, u \rangle - i(\langle u, v \rangle + \langle -iv, v \rangle) \\ &= \|u\|^2 + \overline{\langle -i, 1 \rangle} \langle v, u \rangle - i\langle u, v \rangle - i\langle -i, 1 \rangle \langle v, v \rangle \\ &\stackrel{\text{SLZ}}{=} \|u\|^2 + \|v\|^2 + i\overline{\langle u, v \rangle} - i\langle u, v \rangle \\ &= \|u\|^2 + \|v\|^2 + i(\overline{\langle u, v \rangle} - \langle u, v \rangle) = \|u\|^2 + \|v\|^2 + 2\operatorname{Im}(\langle u, v \rangle) \end{aligned}$$

$$a-bi - (a+bi) = -2bi$$

$$\Rightarrow -2i \operatorname{Im}(z) \quad / : i$$

$$2\operatorname{Im}(z)$$

$$\begin{aligned}
 \|u + \tilde{a}v\|^2 &= \langle u + \tilde{a}v, u + \tilde{a}v \rangle \stackrel{\text{bšl}}{=} \langle u + \tilde{a}v, u \rangle + \langle u + \tilde{a}v, \tilde{a}v \rangle \\
 &= \langle u, u \rangle + \overline{(\tilde{a})} \langle v, u \rangle + \tilde{a} \langle u, v \rangle + \tilde{a} \overline{(\tilde{a})} \langle v, v \rangle \\
 &\stackrel{\text{bšl}}{=} \|u\|^2 + \|v\|^2 + \tilde{a}(\langle u, v \rangle - \overline{\langle u, v \rangle}) = \|u\|^2 + \|v\|^2 - 2\operatorname{Im}(\langle u, v \rangle)
 \end{aligned}$$

$a + b\tilde{a} = a + b\tilde{a} = 2\tilde{a}\operatorname{Im}(b)$

Opet vyžítímme vlastnosti skalárního součinu a posuneme P.45 - $\langle u + v, u \rangle = \langle u, u \rangle + \langle v, u \rangle$

Dále

$$\begin{aligned}
 \|u + \tilde{a}v\|^2 - \|u + iv\|^2 &= \|u\|^2 + \|v\|^2 - 2\operatorname{Im}(\langle u, v \rangle) - \|u\|^2 - \|v\|^2 + 2\operatorname{Im}(\langle u, v \rangle) \\
 &= 4\operatorname{Im}(\langle u, v \rangle), \text{ vydelíme a dostáváme}
 \end{aligned}$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4}(\|u + \tilde{a}v\|^2 - \|u + iv\|^2)$$

■